

PAG-Agent: A biologist-oriented virtual research assistant for context-aware pathway analysis, prioritization, and interpretation

Quang-Huy Nguyen¹, Zeru Zhang¹, Zhandos Sembay², Duc-Hau Le³, Jake Y. Chen², Wei-Shinn Ku¹, Hao Chen¹, Zongliang Yue¹

¹ Auburn University, Auburn, AL, USA ² University of Alabama at Birmingham, Birmingham, AL, USA

³ Hanoi University of Science and Technology, Hanoi, Vietnam

Why PAG-Agent?

• Challenge 1 — Statistical significance ≠ biological relevance

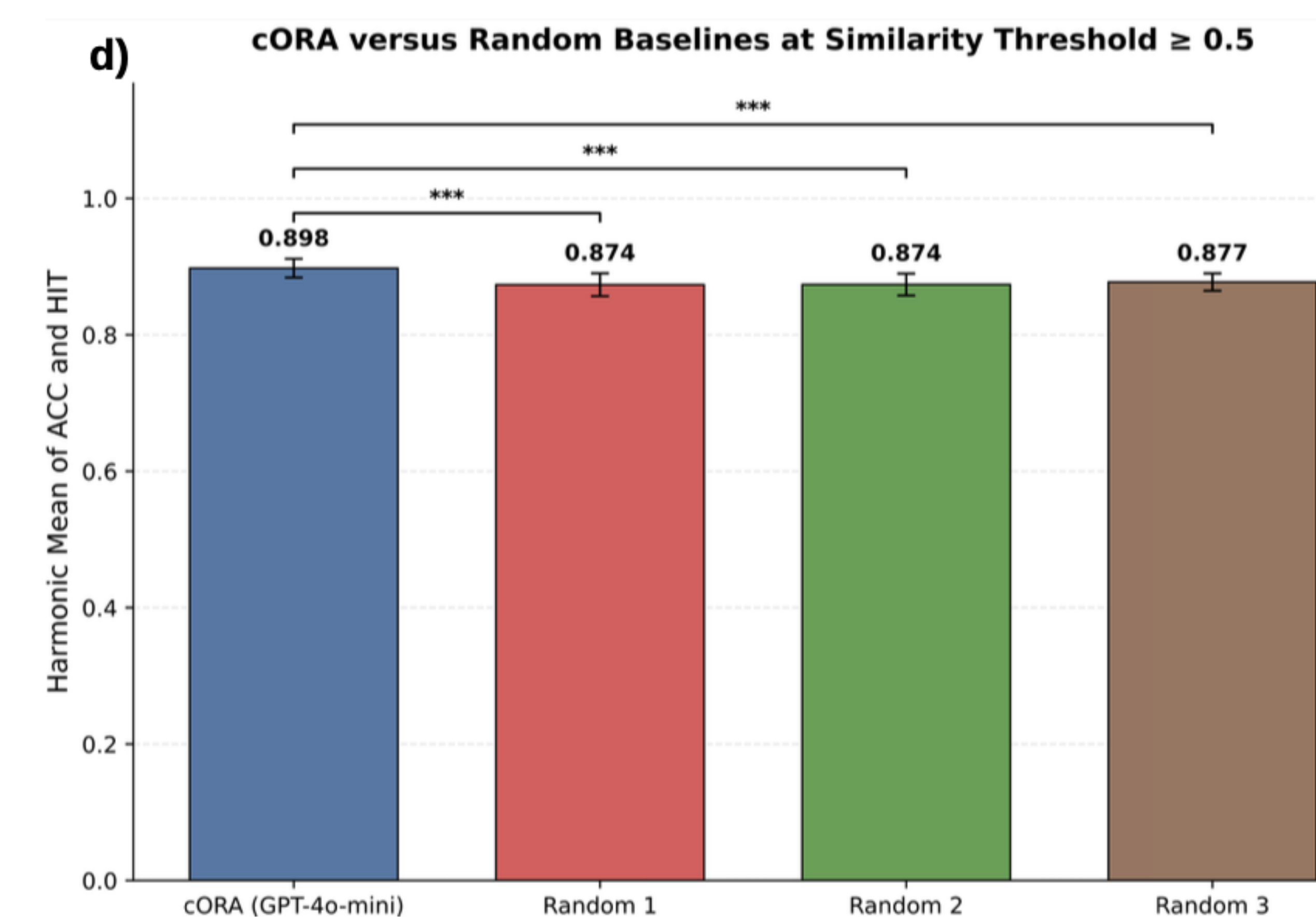
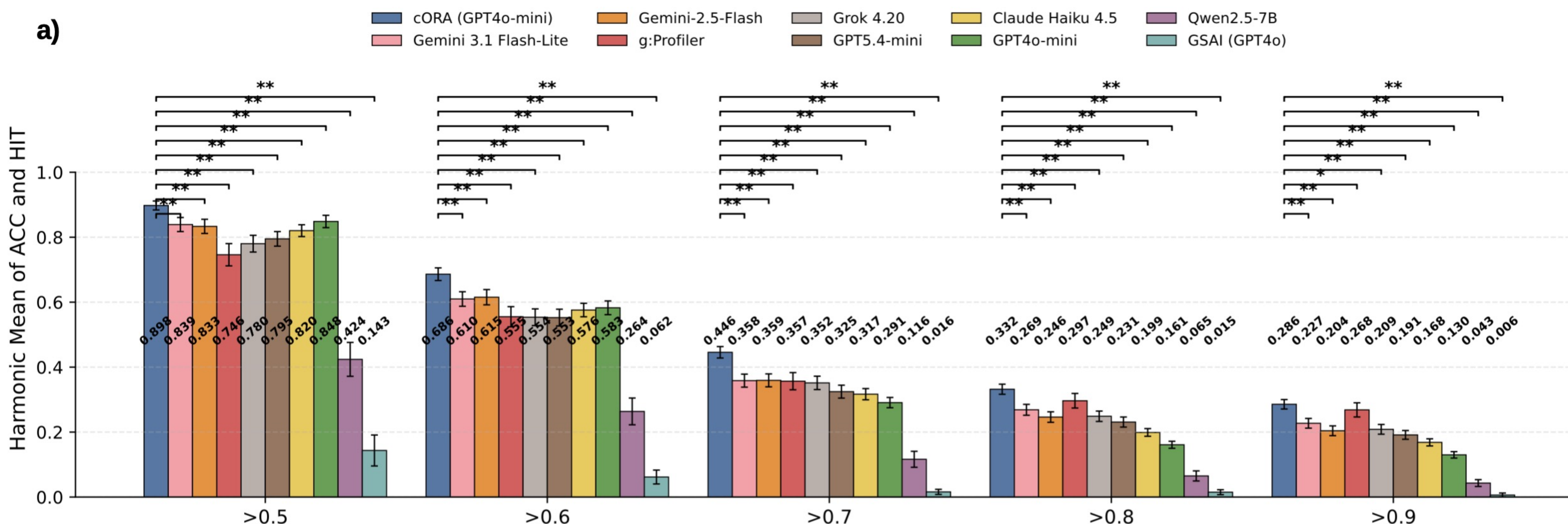
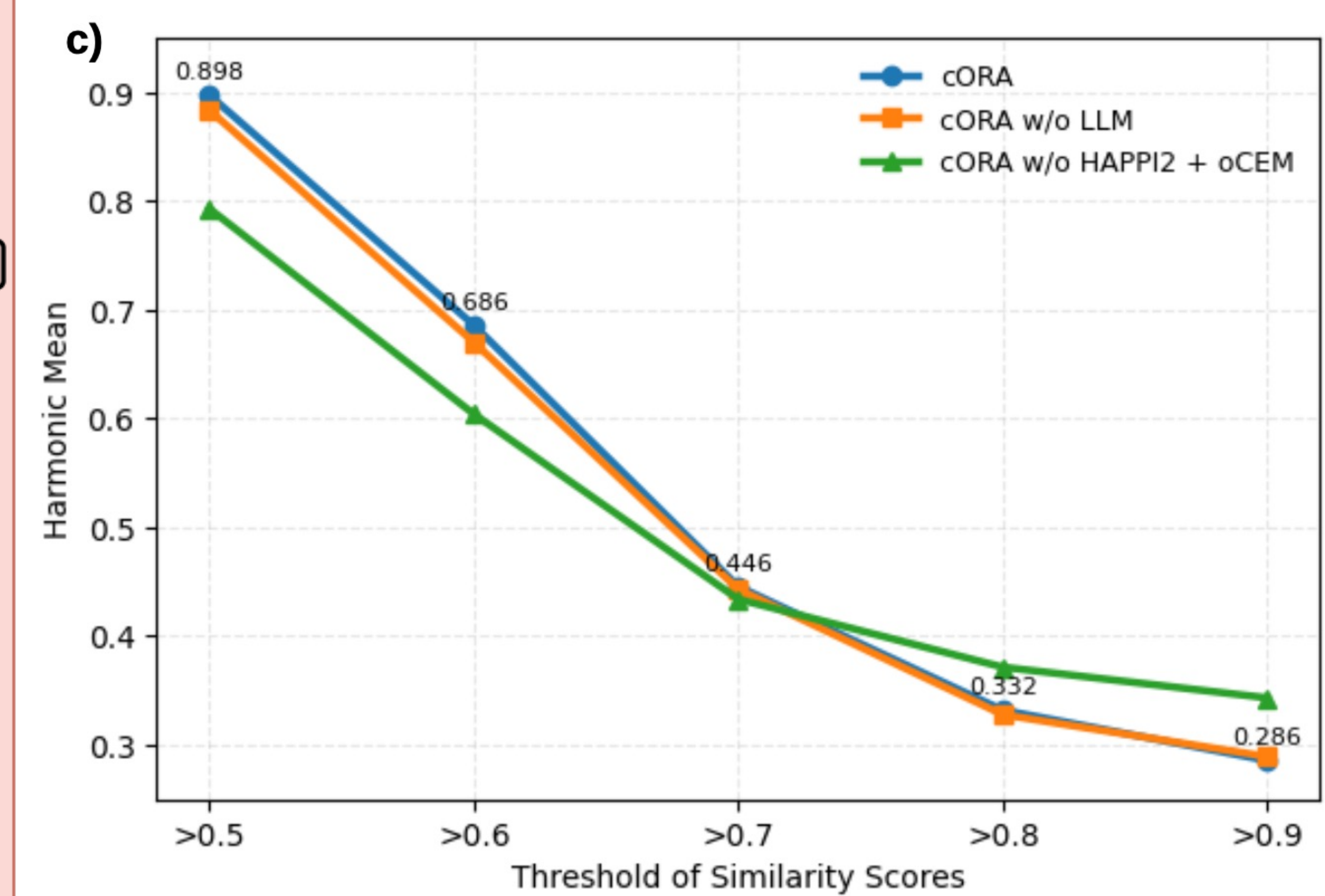
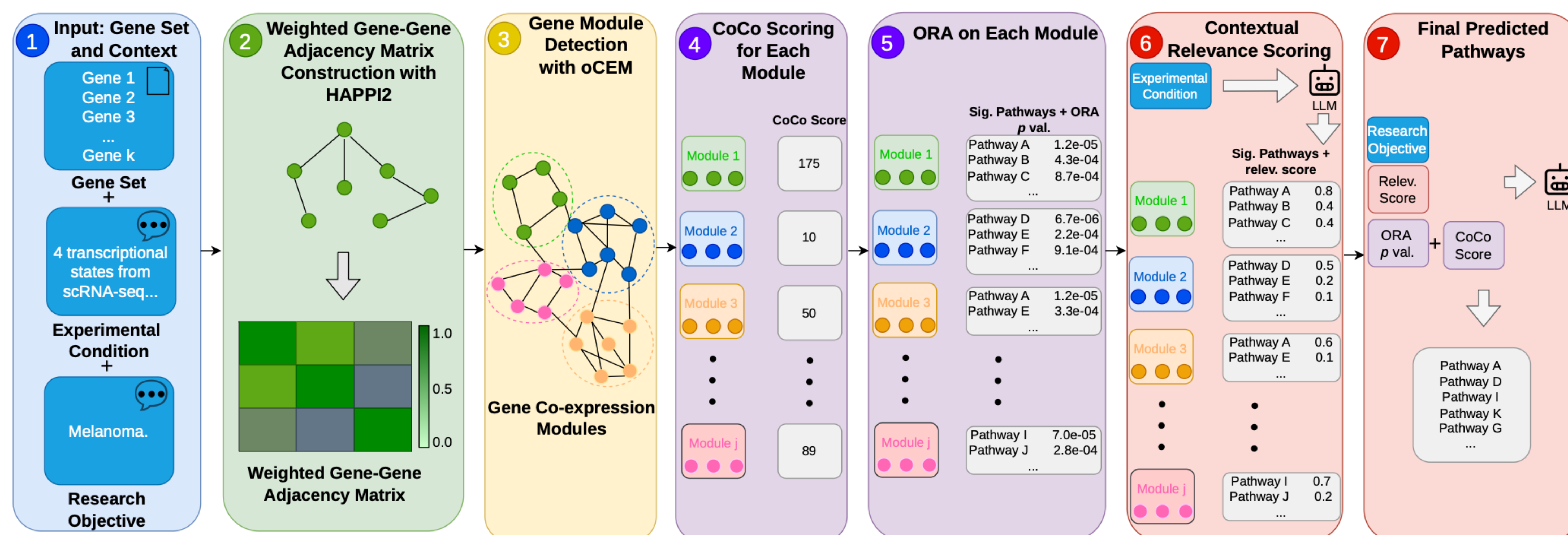
Traditional pathway analysis ranks pathways mainly by significance and does not adequately incorporate the experimental condition or research objective.

• Challenge 2 — Transcriptomic interpretation is fragmented

Researchers move between separate tools for differential expression, pathway analysis, annotation, literature validation, and hypothesis generation.

What is PAG-Agent?

PAG-Agent is a **web-based, biologist-oriented virtual research assistant** that integrates **transcriptomic analysis and biological interpretation** within a **single workflow**. It supports differential expression analysis, pathway analysis, annotation, literature validation, and hypothesis generation through natural-language interaction. Its core innovation, **context-aware Over-Representation Analysis (cORA)**, identifies and prioritizes biologically relevant pathways using **experimental conditions and research objectives**.



Benchmark: 60 cancer-related studies
Each paired with canonical pathways

ACC = 0.913 | HIT = 0.902 | F1= 0.898

Take-home message:

PAG-Agent combines **biologically informed gene modules** with **study-specific context** to prioritize relevant pathways and translate transcriptomic results into **evidence-supported biological insight**.

Test

12:57

please give me relevant citations for associations between Oxidative Phosphorylation and Alzheimer's disease

ChatPathways

12:57

Alzheimer's disease (AD) is a complex neurodegenerative disorder characterized by the accumulation of misfolded proteins and oxidative stress. Several studies have highlighted the association between oxidative phosphorylation and AD pathogenesis.

One key aspect is the role of reactive oxygen species (ROS) in AD progression. ROS production in mitochondria during oxidative phosphorylation is a critical factor in AD pathophysiology [1]. Mitochondrial dysfunction, often observed in AD, contributes to the generation of ROS, which can lead to oxidative stress and subsequent neuronal damage [2].

Moreover, the interplay between proteinopathy and oxidative stress in AD is crucial. Misfolded proteins like beta-amyloid and tau, the hallmarks of AD pathology, are linked to ROS production, further exacerbating oxidative damage in neurons [3]. The harmful trio of "aging, Aβ, and tau protein" has been shown to trigger mitochondrial dysfunction and increase ROS production in AD models [4].

Furthermore, research indicates that elevated ROS levels in AD are associated with the abnormal accumulation of beta-amyloid peptide, tau protein, and heme dyshomeostasis, which in turn are exacerbated by mitochondrial dysfunction and endoplasmic reticulum stress [5].

In summary, the association between oxidative phosphorylation, ROS production, and AD pathogenesis underscores the importance of oxidative stress in the development and progression of Alzheimer's disease.

References:

1. Oxidative Stress, Synaptic Dysfunction, and Alzheimer's Disease. (<https://pmc.ncbi.nlm.nih.gov/articles/PMC5409043/>)

2. The Association of Tau With Mitochondrial Dysfunction in Alzheimer's Disease. (<https://pmc.ncbi.nlm.nih.gov/articles/PMC5874499/>)

3. Linking Oxidative Stress and Proteinopathy in Alzheimer's Disease. (<https://pmc.ncbi.nlm.nih.gov/articles/PMC3898980/>)

4. Insights into mitochondrial dysfunction: aging, amyloid-β, and tau-A. (<https://pubmed.ncbi.nlm.nih.gov/22117646/>)

5. Redox changes and cellular senescence in Alzheimer's disease. (<https://www.sciencedirect.com/science/article/pii/S22132132172400247>)

PAG-Agent Homepage
Research prototype currently hosted on
GitHub Pages